

먼저 피해 서버를 만들기위해 아래코드를 실행 했다.

```
sudo apt update #업데이트
```

```
sudo apt install apache2 -y #아파치2 설치
```

```
sudo systemctl status apache2 #아파치2 상태 확인
```

그리고 먼저 `netstat -ant | grep SYN_RECV` #SYN 로 대기 상태가 있는지를 확인했다.

아무 출력이 없는 것을 확인.

우분투 서버의 ip를 `ip a` 로 192.168.240.132 라는 것을 확인 하였다.

```
ubuntu@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:68:14:72 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altnamex enx000c29681472
    inet 192.168.240.132/24 metric 100 brd 192.168.240.255 scope global dynamic ens33
        valid_lft 1773sec preferred_lft 1773sec
    inet6 fe80::c39f:fe88:1472/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
```

그리고 kali에서 ping test를 위해 ping을 보내보았는데 잘 가는 것을 확인.

```
(kali@kali)-[~]
$ ping 192.168.240.132
PING 192.168.240.132 (192.168.240.132) 56(84) bytes of data.
64 bytes from 192.168.240.132: icmp_seq=1 ttl=64 time=1.05 ms
64 bytes from 192.168.240.132: icmp_seq=2 ttl=64 time=0.619 ms
64 bytes from 192.168.240.132: icmp_seq=3 ttl=64 time=0.652 ms
64 bytes from 192.168.240.132: icmp_seq=4 ttl=64 time=0.659 ms
64 bytes from 192.168.240.132: icmp_seq=5 ttl=64 time=0.642 ms
64 bytes from 192.168.240.132: icmp_seq=6 ttl=64 time=0.631 ms
64 bytes from 192.168.240.132: icmp_seq=7 ttl=64 time=0.903 ms
64 bytes from 192.168.240.132: icmp_seq=8 ttl=64 time=1.09 ms
64 bytes from 192.168.240.132: icmp_seq=9 ttl=64 time=0.786 ms
64 bytes from 192.168.240.132: icmp_seq=10 ttl=64 time=0.974 ms
64 bytes from 192.168.240.132: icmp_seq=11 ttl=64 time=1.04 ms
64 bytes from 192.168.240.132: icmp_seq=12 ttl=64 time=1.12 ms
^C
— 192.168.240.132 ping statistics —
12 packets transmitted, 12 received, 0% packet loss, time 11097ms
rtt min/avg/max/mdev = 0.619/0.846/1.116/0.192 ms
```

그 다음 칼리에서 우분투 ip 공격 `sudo hping3 -S --flood -V -p 80 192.168.240.128`

```
(kali㉿kali)-[~]
└─$ sudo hping3 -S --flood -V -p 80 192.168.240.132
using eth0, addr: 192.168.240.128, MTU: 1500
HPING 192.168.240.132 (eth0 192.168.240.132): S set, 40 headers + 0 data byte
s
hping in flood mode, no replies will be shown

^C
— 192.168.240.132 hping statistic —
59325989 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

다음 우분투에서 `netstat -ant | grep SYN_RECV` 해당 코드로 상태를 다시 확인 해 보았다.

```
tcp6      0      0 192.168.240.132:80    192.168.240.128:38693 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:42170 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:2682  SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:44350 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:54995 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:41769 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:31167 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:45191 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:50802 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:59394 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:63905 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:46865 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:1329  SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:16361 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:15730 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:49901 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:64124 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:51095 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:25437 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:9913  SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:47435 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:63926 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:56722 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:24586 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:24447 SYN_RECV
tcp6      0      0 192.168.240.132:80    192.168.240.128:65177 SYN_RECV
ubuntu@ubuntu:~$
```

수 많은 SYN_RECV를 확인할 수 있다.

`sysctl net.ipv4.tcp_syncookies` 해당 코드로 쿠키 값 확인.

`net.ipv4.tcp_syncookies = 1` 1개가 출력 되는 것을 확인할 수 있다.

다음 `sudo sysctl -w net.ipv4.tcp_syncookies=1` 코드로 방어 시도.

```
ubuntu@ubuntu:~$ sysctl net.ipv4.tcp_syncookies
net.ipv4.tcp_syncookies = 1
ubuntu@ubuntu:~$ sudo sysctl -w net.ipv4.tcp_syncookies=1
[sudo: authenticate] Password:
net.ipv4.tcp_syncookies = 1
```

그리고 칼리에서 다시 공격해 보았다

```
(kali㉿kali)-[~]  
$ sudo hping3 -S --flood -V -p 80 192.168.240.132  
using eth0, addr: 192.168.240.128, MTU: 1500  
HPING 192.168.240.132 (eth0 192.168.240.132): S set, 40 headers + 0 data byte  
s  
hping in flood mode, no replies will be shown  
^C  
— 192.168.240.132 hping statistic —  
20233055 packets transmitted, 0 packets received, 100% packet loss  
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

다시 SYN_RECV 대기 상태 조회.

```
ubuntu@ubuntu:~$ sudo sysctl -w net.ipv4.tcp_syncookies=1  
net.ipv4.tcp_syncookies = 1  
ubuntu@ubuntu:~$ netstat -ant | grep SYN_RECV  
ubuntu@ubuntu:~$  
ubuntu@ubuntu:~$
```

하나도 없는 것을 확인할 수 있다.

방어성공.